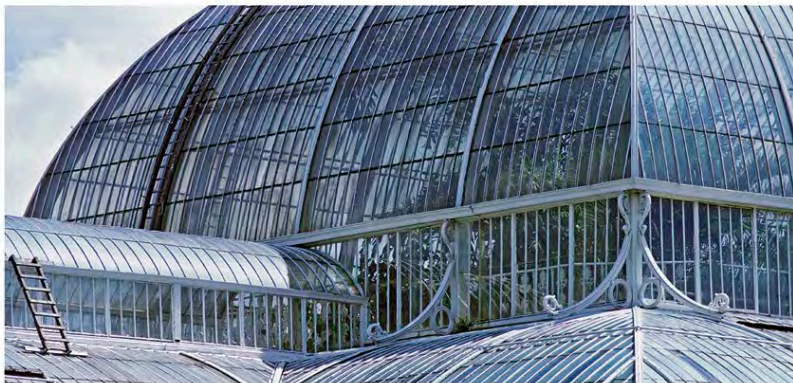


IRON, STEEL, AND BEYOND

For over 10,000 years, the basic materials for construction were earth, timber, bricks, and stone. In the last 200 years, however, there has been radical change. Our architecture is now given new and ever greater freedoms of form through materials such as iron, steel, and concrete.

Advent of new materials

Not only did the Industrial Revolution in the 18th–19th centuries herald the advent of industry and new forms of transportation, but it changed the way buildings were made. Engineers were quick to see the possibilities in the large-scale production of iron. The metal was fire-resistant and could be molded into shapes that were impossible to make with stone or brick. It was first tested out in 1779, with the Iron Bridge over the River Severn in England, and was then adapted for columns, beams, and full iron-frame buildings elsewhere. Ditherington Mill, built in 1796 in Shrewsbury, England, is considered the first iron-framed building in the world. In the late 19th century, iron was superseded by steel, which was much tougher and less brittle. Its qualities were shown off to best advantage in 1889, when steel was the main building material for the revolutionary Eiffel Tower (see pp.180–181).



△ MAIDEN USE

One of the earliest uses of the full iron frame was in the building of greenhouses, such as the Palm House at Kew Gardens, England.

▷ FREE-FLOWING STRUCTURE

Supported by 16 inward-curving columns, Oscar Niemeyer's Cathedral of Brasilia (see pp.84–85) is an example of the freedom of structure offered to architecture by the innovation of prestressed reinforced concrete.

